

Assignment No. 01: Algorithm Writing Styles

BSSE 5B

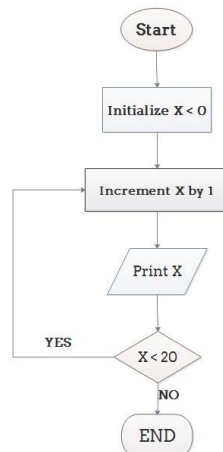
Fall 2025

Due Date: October 12, 2025

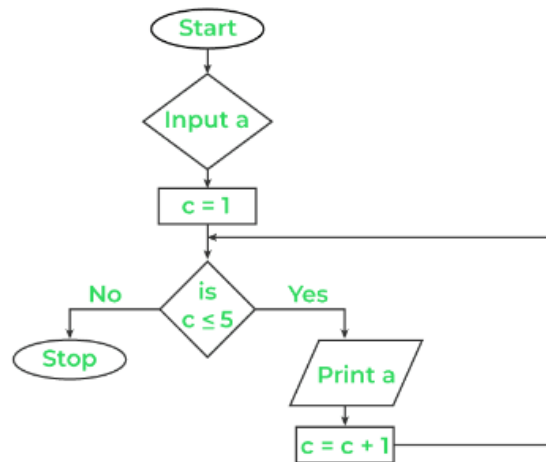
Instructions:

Answer all questions clearly and concisely. For algorithm-related tasks, include both pseudocode and, if applicable, a flowchart. Submit your answers in PDF format.

- Q. 1:** Explain what pseudocode is and why it is used. How does it differ from a formal programming language?
- Q. 2:** List and explain at least three advantages of using pseudocode to describe algorithms before coding.
- Q. 3:** Compare and contrast pseudocode and flowcharts. In what scenarios would you prefer one over the other?
- Q. 4:** Write pseudocode for a **linear search algorithm** that finds a specific value in an array of integers. Clearly label the input, process, and output.
- Q. 5:** Draw a flowchart for the **linear search algorithm** described in question 4. Include all relevant symbols to depict input, output, and decision-making.
- Q. 6:** Write pseudocode that checks whether a given number is positive, negative, or zero, and prints the corresponding message.
- Q. 7:** Design a flowchart for an algorithm that sums all numbers from 1 to 100. Include decision-making for the looping process.
- Q. 8:** Below is a flowchart: Write the pseudocode that corresponds to this flowchart.



Q. 9: Below is a flowchart. Write the pseudocode that corresponds to this flowchart.



Q. 10: Write pseudocode for a **bubble sort algorithm** that sorts a list of numbers in ascending order. Explain each step in your pseudocode.

Q. 11: Discuss the strengths and weaknesses of both pseudocode and flowcharts in algorithm design. In your opinion, which is more useful for beginners and why?